

Week 3 Practical

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PART 1: OPEN MONGODB ATLAS

MongoDB Atlas

ORGANIZATION: NAVADEEP V/PPARTH's Org

All Projects

Find a project...

Project Name	Clusters	Tags	Users	Teams	Alerts	Actions
Project 0	1 Cluster	+ Add Tags	1 User	0 Teams	0 Alerts	

Items per page: 25

1 - 1 of 1 items

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PART 2: CREATE book_metadata COLLECTION

MongoDB Atlas

ORGANIZATION: NAVADEEP V/PPARTH's Org PROJECT: Project 0

Data Explorer

Cluster0

View monitoring Visualize your data Create database Refresh

Database name	Storage size	Data size	Collections	Indexes
MongoDB	73.73 kB	3.29 kB	2	2
admin	0 B	0 B	0	0
bookflow_db	40.96 kB	2.14 kB	2	2
local	-	-	0	-
sample_mflix	136.13 MB	129.12 MB	6	10
test	36.86 kB	99.00 B	1	1

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PART 3: DOCUMENT MODELING

ORGANIZATION
NAVADEEP VIPPART...

PROJECT
Project 0

Cluster0

Cluster0

View monitoring

Visualize your data

Create database

Refresh

Database name	Storage size	Data size	Collections	Indexes
MongoDB	73.73 kB	3.29 kB	2	2
admin	0 B	0 B	0	0
bookflow_db	40.96 kB	2.14 kB	2	2
local	-	-	0	-
sample_mflix	136.13 MB	129.12 MB	6	10
test	36.86 kB	99.00 B	1	1

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PART 4: INSERT SAMPLE DOCUMENTS

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PROJECT
Project 0

book_metadata

Cluster0 > bookflow_db > book_metadata

View monitoring

Visualize Your Data

Documents

Aggregations

Schema

Indexes

Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA

BULK

EXPORT CODE

25 1-5 of 5

	book_metadata._id ObjectId	book_id Int32	title String	author String	published_year Int32	
1	ObjectId('6a957a8ca519d4...')	101	"The Martian"	"Andy Weir"	2021	
2	ObjectId('6a957a30a519d4...')	102	"Dune"	"Frank Herbert"	2022	
3	ObjectId('6a957a3ba519d4...')	103	"Clean Code"	"Robert C. Martin"	2019	
4	ObjectId('6a957a47a519d4...')	104	"Artificial Intelligence..."	"John Smith"	2023	
5	ObjectId('6a957a56a519d4...')	105	"Data Structures"	"Mark Allen"	2024	

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PART 5: VERIFY INSERTED DOCUMENTS

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Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- MongoDB
- admin
- bookflow_db
 - book_metadata**
 - books
 - local
 - sample_mflix
 - test

Cluster0 > bookflow_db > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

[View monitoring](#) [Visualize Your Data](#)

[ADD DATA](#) [BULK](#) [EXPORT CODE](#)

25 1-5 of 5

	book_metadata_id ObjectId	book_id Int32	title String	author String	published_year Int32	
1	ObjectId('6a957a8ca519d4...')	101	"The Martian"	"Andy Weir"	2021	
2	ObjectId('6a957a38a519d4...')	102	"Dune"	"Frank Herbert"	2022	
3	ObjectId('6a957a3ba519d4...')	103	"Clean Code"	"Robert C. Martin"	2019	
4	ObjectId('6a957a47a519d4...')	104	"Artificial Intelligence..."	"John Smith"	2023	
5	ObjectId('6a957a56a519d4...')	105	"Data Structures"	"Mark Allen"	2024	

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PART 6: COMPARISON OPERATOR \$gt

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Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- MongoDB
- admin
- bookflow_db
 - book_metadata**
 - books
 - local
 - sample_mflix
 - test

Cluster0 > bookflow_db > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

[View monitoring](#) [Visualize Your Data](#)

[Generate query](#) [Explain](#) [Reset](#) [Find](#) [Options](#)

[ADD DATA](#) [BULK](#) [EXPORT CODE](#)

25 1-5 of 5

```
{
  "reviews.rating": {
    $gt: 4
  }
}
```

	itring	file_size String	member_id Int32	reviews Array	dimensions Object	
1	PDF"	"2.5 MB"	101	[] 3 elements	No field	
2	EPUB"	"3.1 MB"	102	[] 3 elements	No field	
3	l"	No field	103	[] 2 elements	{} 3 fields	
4	PDF"	"4.2 MB"	104	[] 3 elements	No field	
5	PDF"	"5.0 MB"	105	[] 2 elements	No field	

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PART 7: USING \$IN

ORGANIZATION: NAVADEEP VIPPART... PROJECT: Project 0

Data Explorer

My Queries | Data Modeling

CLUSTERS (1)

Cluster0

- MongoDB
- admin
- bookflow_db
 - book_metadata
 - books
 - local
 - sample_mflix
 - test

Cluster0 > bookflow_db > book_metadata

Documents | Aggregations | Schema | Indexes (1) | Validation

Generate query * Explain Reset Find Options ▶

25 1 - 5 of 5

	book_metadata_id ObjectId	book_id Int32	title String	author String	published_year Int
1	ObjectId('6a957c93c396b4...')	101	"The Martian"	"Andy Weir"	2021
2	ObjectId('6a957c93c396b4...')	102	"Dune"	"Frank Herbert"	2022
3	ObjectId('6a957c93c396b4...')	103	"Clean Code"	"Robert C. Martin"	2019
4	ObjectId('6a957c93c396b4...')	104	"Artificial Intelligence..."	"John Smith"	2023
5	ObjectId('6a957c93c396b4...')	105	"Data Structures"	"Mark Allen"	2024

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PART 8: REGEX QUERY

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Data Explorer

My Queries | Data Modeling

CLUSTERS (1)

Cluster0

- MongoDB
- admin
- bookflow_db
 - book_metadata
 - books
 - local
 - sample_mflix
 - test

Cluster0 > bookflow_db > book_metadata

Documents | Aggregations | Schema | Indexes (1) | Validation

Generate query * Explain Reset Find Options ▶

25 1 - 4 of 4

	book_metadata_id ObjectId	book_id Int32	title String	author String	published_year Int32
1	ObjectId('6a957c93c396b4...')	101	"The Martian"	"Andy Weir"	2021
2	ObjectId('6a957c93c396b4...')	102	"Dune"	"Frank Herbert"	2022
3	ObjectId('6a957c93c396b4...')	104	"Artificial Intelligence..."	"John Smith"	2023
4	ObjectId('6a957c93c396b4...')	105	"Data Structures"	"Mark Allen"	2024

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PART 9: AGGREGATION PIPELINE

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My Queries
Data Modeling

CLUSTERS (1)
Search clusters

Cluster0
MongoDB
admin
bookflow_db
book_metadata
books
local
sample_mflix
test

book_metadata

Cluster0 > bookflow_db > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Stage 1 \$match

Output preview after \$match stage (Sample of 4 documents)

Generate aggregation Explain Run Options

PREVIEW STAGES TEXT WIZARD

1 {
2 "published_year": {
3 "\$gt": 2020
4 }
5 }
6 }

1: ObjectId('6a957c93c396b49046216614')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101

1: ObjectId('6a957c93c396b49046216614')
book_id: 102
title: "Dune"
author: "Frank Herbert"
published_year: 2022
category: "Science Fiction"
format: "Digital EPUB"
file_size: "3.1 MB"
member_id: 102

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My Queries
Data Modeling

CLUSTERS (1)
Search clusters

Cluster0
MongoDB
admin
bookflow_db
book_metadata
books
local
sample_mflix
test

book_metadata

Cluster0 > bookflow_db > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Stage 2 \$unwind

Output preview after \$unwind stage (Sample of 10 documents)

Generate aggregation Explain Run Options

PREVIEW STAGES TEXT WIZARD

1 "\$reviews"

1: ObjectId('6a957c93c396b49046216614')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101

1: ObjectId('6a957c93c396b49046216614')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101

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My Queries
Data Modeling

CLUSTERS (1)
Search clusters

Cluster0
MongoDB
admin
bookflow_db
book_metadata
books
local
sample_mflix
test

book_metadata

Cluster0 > bookflow_db > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Stage 3 \$group

Output preview after \$group stage (Sample of 4 documents)

Generate aggregation Explain Run Options

PREVIEW STAGES TEXT WIZARD

1 {
2 "_id": "\$title",
3 "average_rating": {
4 "\$avg": "\$reviews.rating"
5 },
6 "total_reviews": {
7 "\$sum": 1
8 }
9 }

1: "The Martian"
average_rating: 4.666666666666667
total_reviews: 3

1: "Artificial Intelligence Basics"
average_rating: 4.333333333333333
total_reviews: 3

ORGANIZATION NAVADEEP VIPPART... PROJECT Project 0

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- MongoDB
- admin
- bookflow_db
 - book_metadata
 - books
- local
- sample_mflix
- test

Cluster0 > bookflow_db > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Smatch Sunwind Sgroup Sort

Generate aggregation Explain Run Options

Untitled - modified SAVE CREATE NEW EXPORT CODE

PREVIEW STAGES TEXT WIZARD

Stage 4 \$sort

1 * {
2 "average_rating": -1
3 }

Output preview after \$sort stage (Sample of 4 documents)

_id: "The Martian"
average_rating: 4.666666666666667
total_reviews: 3

_id: "Dune"
average_rating: 4.666666666666667
total_reviews: 3

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Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- MongoDB
- admin
- bookflow_db
 - book_metadata
 - books
- local
- sample_mflix
- test

Cluster0 > bookflow_db > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Smatch Sunwind Sgroup Sort Edit

Explain Run Options

ALL RESULTS EXPORT CODE

Showing 1 - 4 count results

_id: "The Martian"
average_rating: 4.666666666666667
total_reviews: 3

_id: "Dune"
average_rating: 4.666666666666667
total_reviews: 3

_id: "Artificial Intelligence Basics"
average_rating: 4.333333333333333
total_reviews: 3

_id: "Data Structures"
average_rating: 4
total_reviews: 2

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PART 10: CREATE TEXT INDEX

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Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- MongoDB
- admin
- bookflow_db
 - book_metadata
 - books
- local
- sample_mflix
- test

book_metadata

Cluster0 > bookflow_db > book_metadata

Documents 5 Aggregations Schema **Indexes 2** Validation

Create Index Refresh

Looking for search indexes?
These indexes can be created and viewed under Search and Vector Search.

Name & Definition	Type	Size	Usage	Properties	Status
id	REGULAR	36.9 kB	2 (since Mon Aug 31 2026)	UNIQUE	READY
reviews.comments_text	TEXT	20.5 kB	0 (since Mon Aug 31 2026)		READY

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https://cloud.mongodb.com/v2/6a7eb8811876dbac38db357?utm_source=DE&utm...

PART 11: SEARCH USING TEXT INDEX

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Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- MongoDB
- admin
- bookflow_db
 - book_metadata
 - books
- local
- sample_mflix
- test

book_metadata

Cluster0 > bookflow_db > book_metadata

Documents 5 Aggregations Schema **Indexes 2** Validation

Generate query

```
{
  "$text": {
    "$search": "excellent"
  }
}
```

ADD DATA BULK EXPORT CODE

25 1-5 of 5

	_id ObjectId	book_id Int32	title String	author String	published_year Int32	category String
1	ObjectId('6a957c93c396b4...')	181	"The Martian"	"Andy Weir"	2011	"Science Fiction"
2	ObjectId('6a957c93c396b4...')	183	"Clean Code"	"Robert C. Martin"	2019	"Programming"
3	ObjectId('6a957c93c396b4...')	182	"Dune"	"Frank Herbert"	2022	"Science Fiction"
4	ObjectId('6a957c93c396b4...')	185	"Data Structures"	"Mark Allen"	2024	"Computer Science"
5	ObjectId('6a957c93c396b4...')	184	"Artificial Intelligence..."	"John Smith"	2023	"Artificial Intelli..."

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PART 12: SEARCH FOR A SPECIFIC KEYWORD

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Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - MongoDB
 - admin
 - bookflow_db
 - book_metadata

Cluster0 > bookflow_db > book_metadata

Documents 5 Aggregations Schema Indexes 2 Validation

```
{ "$text": { "$search": "\"science fiction\"" } }
```

Generate query Explain Reset Find Options

ADD DATA BULK EXPORT CODE

25 1 - 2 of 2

	book_metadata	_id ObjectId	book_id Int32	title String	author String	published_year Int32	category String
1		ObjectId('6a957c93c396b4...')	182	"Dune"	"Frank Herbert"	2022	"Science Fiction"
2		ObjectId('6a957c93c396b4...')	181	"The Martian"	"Andy Weir"	2021	"Science Fiction"

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PART 13: USE explain()

Explain Plan

Explain provides key execution metrics that help diagnose slow queries and optimize index usage. [Learn more](#)

Visual Tree Row Output

```
graph TD; IXSCAN[IXSCAN] --> FETCH[FETCH]; FETCH --> TEXT_MATCH[TEXT_MATCH]
```

TEXT_MATCH
Returned 5 Execution Time 0 ms

FETCH
Returned 5 Execution Time 0 ms

IXSCAN
Returned 5 Execution Time 0 ms
Index Name: reviews.comment...
Multi Key Index: yes

Query Performance Summary

- 5 documents returned
- 5 documents examined
- 0 ms execution time
- Is not sorted in memory
- 5 index keys examined

Query used the following index:
_fts (text)
_ftsx ↑

Close

PART 14: DATA EXPLORER GUI METHOD

ORGANIZATION
NAVADEEP VIPPART...

PROJECT
Project 0

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- MongoDB
- admin
- bookflow_db
 - book_metadata
 - books
- local
- sample_mflix
- test

book_metadata

Cluster0 > bookflow_db > book_metadata

Documents 5 Aggregations Schema **Indexes 2** Validation

Create Index Refresh

Looking for search indexes?
These indexes can be created and viewed under [Search and Vector Search](#).

Name & Definition	Type	Size	Usage	Properties	Status
> _id_	REGULAR	36.9 kB	2 (since Mon Aug 31 2026)	UNIQUE	READY
> reviews.comments_text	TEXT	20.5 kB	6 (since Mon Aug 31 2026)		READY

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